The leading coefficient obstruction to an Eisenstein generator

Conjecture 00000003898

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Fix a prime p. Write $\mathbb{Z}_p$ for the p-adic integers and $v_p(p) = 1$ for the normalized valuation. A polynomial $E \in \mathbb{Z}_p[q]$ is Eisenstein at (p) only if its leading coefficient does not belong to (p); all lower coefficients belong to (p) and the constant coefficient does not belong to $(p)^2$.

Theorem 1. *There is no polynomial $E \in \mathbb{Z}_p[q]$ that is Eisenstein at (p) and whose leading coefficient has valuation one.*

Proof. Let a be its leading coefficient. The valuation condition $v_p(a) = 1$ implies $a \neq 0$ and $a \in (p)$. Eisenstein's leading-coefficient condition says $a \notin (p)$. These contradict each other. □

This disproves the existence assertion about the generator before any property of the map θ is needed. Requiring the polynomial also to generate a kernel or characterize an admissible parameter cannot make a nonexistent polynomial exist. In the usual definition it is the *constant* coefficient of an Eisenstein polynomial that has valuation exactly one; the leading coefficient is a unit. Changing "leading" to "constant" would be a different problem. Both official language versions say leading coefficient.

Lean correspondence. The project uses Mathlib's p-adic integers, Eisenstein predicate, ideal span and valuation. The library theorem `mem_span_pow_iff_le_valuation` converts the valuation condition into ideal membership. `leading_valuation_ne_one` proves the contradiction for every prime p and polynomial; `conjecture_false` negates the required existential statement. The type `Fact p.Prime` expresses the prime hypothesis and is not a custom axiom. No prism construction or conjectured theorem is assumed.

Verification. Lean 4.19.0 with Mathlib pinned to `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The accompanying project and verification transcript give the exact theorem and axiom audit. No proof placeholders, native evaluator proof bypass, or custom axiom declarations are used. This is a prepared submission; official acceptance has not been claimed.